

DSA4213 Final Project Proposal

Zhang Zhenjie
e0957630@u.nus.edu
A0252042W

September 16, 2025

1 Introduction

Modern e-commerce and social media platforms rely heavily on information retrieval (IR) and recommendation systems to deliver relevant results to users. Recently, Retrieval-Augmented Generation (RAG) has recently gained traction as a way to enhance large language model (LLM) outputs by grounding them in retrieved documents. However, most RAG systems rely on a fixed retrieval strategy such as keyword search, dense embeddings, or hybrid methods. This rigidity often results in suboptimal retrieval for diverse query types.

This project aims to address the problem of adaptive retrieval: how can we dynamically select the best retrieval strategy for a given query? Inspired by online learning in recommender systems, I propose evaluating bandit algorithms in a simulated retrieval setting. Using the MS MARCO dataset, we will treat candidate retrievers as “arms” and study which algorithms converge fastest and yield the highest cumulative reward. The emphasis will be on ablation studies across bandit methods rather than building a full RAG pipeline, due to time constraints and the scope of a solo project.

This problem excites me for three reasons. First, adaptive retrieval has direct industrial relevance in e-commerce and social media search systems. Second, multi-armed bandits belong to the field of online reinforcement learning, offering a contrast to traditional supervised deep learning approaches. Finally, retrieval quality is foundational to RAG, poor retrieval inevitably leads to poor generation.

2 Model

2.1 Dataset

We will use the *MS MARCO* dataset, which is explicitly designed for retrieval and QA tasks, and provides relevance labels suitable for automated evaluation.

2.2 Methods

The objective is to investigate whether online learning algorithms like Multi-Armed Bandits can dynamically and effectively select the best retrieval method (BM25, dense vector, keyword, or hybrid) for individual queries.

We will explore three bandit algorithms: ϵ -greedy, Upper Confidence Bound (UCB), and Thompson Sampling.

The setup is as follows:

1. Candidate retrieval methods are treated as the “arms” of the bandit.
2. For each incoming query, the bandit algorithm selects a retriever.
3. The chosen retriever fetches the top- k documents.
4. A reward is assigned based on relevance (e.g., reward = 1 if at least one of the top-3 documents is relevant according to ground truth, else 0).
5. The bandit updates its policy accordingly.

The final output is not generated text, but (i) a superior set of retrieved documents, and (ii) a learned policy for choosing the right retriever for different query types.

3 Expected Outcome

The primary evaluation metric will be cumulative reward (or equivalently, cumulative regret) over time, plotted against static baselines to measure learning performance.

Secondary metrics will include information retrieval measures such as nDCG@k, Precision@k, and Recall@k, comparing the fully trained adaptive system against static baselines. This allows us to evaluate retrieval quality without relying on expensive human or LLM-based judgments.

We will also analyze convergence speed, stability, and robustness to noise. Additionally, qualitative examples will be presented, showing cases where the bandit correctly learns to prefer BM25 (e.g., keyword-heavy, product-specific queries) versus dense retrieval (e.g., semantic or verbose queries).

Some ablation studies that we can do includes varying ε for ε -greedy, confidence multiplier for UCB, and different priors for Thompson Sampling. We can also vary the size of retrieval candidate set passed to bandit such as top-10, top-25, top-50 which would affect information retrieval metrics such as nDCG@k.

4 Future Work

While the proposed project focuses on standard Multi-Armed Bandit algorithms, a natural extension is to explore **Contextual Bandits** if time permits. Unlike basic bandits, which treat all queries identically, contextual bandits leverage side information (e.g., query features such as length, keyword density, or even dense embeddings) to inform arm selection. This would allow the retrieval strategy to adapt not just over time, but also per query instance.

An even more advanced extension is to apply **Neural Contextual Bandits**, where a neural network models the expected reward given a query embedding and candidate retriever. This approach is particularly appealing for high-dimensional contexts and could capture nonlinear relationships between query characteristics and retrieval effectiveness. However, due to implementation complexity, this direction is better suited as future work beyond the scope of the current project.